

Final Report

HOTEL BOOKING ANALYSIS

ABSTRACT

The Hotel Booking Analysis project focuses on analyzing hotel booking data to understand customer behavior, booking patterns, cancellation trends, and revenue generation. The project uses Excel for data cleaning, MySQL for data analysis, and Power BI for creating interactive dashboards. The analysis helps in improving customer retention and supports better business decision-making.

INTRODUCTION

The travel and hospitality industry generates a large amount of booking data every day. Analyzing this data helps hotels understand customer preferences and improve business performance. This project analyzes hotel booking information to identify trends and generate meaningful business insights.

OBJECTIVES

- Analyze hotel booking trends
- Study customer behavior
- Identify cancellation patterns
- Generate revenue insights
- Create an interactive dashboard
- Support business decision-making

TOOLS USED

- Excel – Data Cleaning
- MySQL – Data Analysis
- Power BI – Dashboard Visualization
- GitHub – Project Storage

DATASET DESCRIPTION

The Hotel Booking Dataset was downloaded from Kaggle. The dataset contains booking information such as hotel type, lead time, arrival date, customer type, country, average daily rate (ADR), cancellation status, and reservation details.

METHODOLOGY

Step 1: Data Collection

The hotel booking dataset was downloaded from Kaggle.

Step 2: Data Cleaning

- Removed duplicate values
- Handled missing values
- Formatted date fields
- Checked data consistency

Step 3: SQL Analysis

- SQL queries were used for:
- Total bookings analysis
- Revenue analysis
- Customer type analysis
- Cancellation analysis
- Country-wise booking analysis

Step 4: Dashboard Development

- Power BI dashboard was created using:
- KPI Cards
- Revenue charts
- Booking trends
- Customer analysis
- Interactive filters and slicers

DASHBOARD FEATURES

- Total Bookings KPI
- Revenue KPI
- Average Price KPI
- Cancellation KPI
- Booking Trend Analysis
- Country Analysis
- Customer Analysis
- Interactive Filters

BUSINESS INSIGHTS

- City hotels generate higher revenue.
- Customer behavior influences booking trends.
- High lead time increases cancellation rates.
- Seasonal trends impact hotel bookings.
- Dynamic pricing helps improve revenue performance.

CONCLUSION

The Hotel Booking Analysis project successfully analyzed hotel booking data and generated useful business insights. SQL and Power BI were used to visualize trends and understand customer behavior. The dashboard supports data-driven decision-making and helps improve business strategies.

FUTURE ENHANCEMENTS

- Apply machine learning techniques
- Implement customer segmentation
- Add predictive analytics